

Fig. 3: Actual-size PCB of interval timer

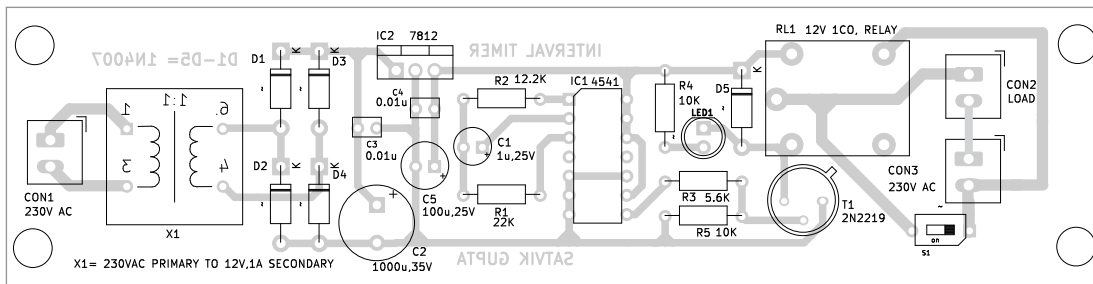


Fig. 4: Component layout of the PCB

PARTS LIST

Semiconductors:

- IC1 - LM7812, 12V voltage regulator
- IC2 - CD4541BE timer
- D1-D5 - 1N4007 rectifier diode
- LED1 - 5mm LED
- T1 - 2N2219 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 22-kilo-ohm
- R2 - 12.2-kilo-ohm
- R3 - 5.6-kilo-ohm
- R4, R5 - 10-kilo-ohm

Capacitors:

- C1 - 1 μ F, 16V electrolytic
- C2 - 1000 μ F, 25V electrolytic
- C3, C4 - 0.01 μ F ceramic disk
- C5 - 100 μ F, 25V electrolytic

Miscellaneous:

- CON1-CON3 - 2-pin connector
- S1 - 6 Amps on/off switch
- RL1 - 12V, 1C/O relay
- X1 - 230V AC primary to 12V, 1A secondary transformer

DC regulated power supply from the input 230V AC supply through the stepdown transformer. When you connect 230V AC to CON1, the circuit becomes enabled and pin 8 of IC2 goes high and the relay energises. The connected load (either fan or light) gets supply from CON3 through the N/O (normally open) contact of RL1. After 15 minutes IC

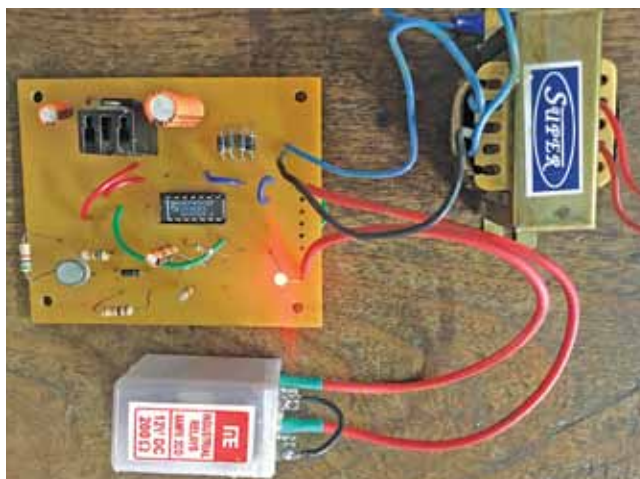


Fig. 5: Author's prototype of interval timer

output pin 8 will go low, de-energising the relay RL1 and the load.

To switch on the device again, you must switch off (disconnect 230V AC from CON1) and switch it on to get another preset time usage.

A bypass switch (S1) is provided across the relay contacts, which should normally be kept off. In case, continuous running of the fan or light is required, the bypass switch can be turned to its on position. An LED1 is provided across the relay coil to

indicate the timer operation.

Construction and testing

An actual-size, single-side PCB layout for the interval timer is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable box.

Alternatively, you can assemble the circuit on a general-purpose PCB, as shown in author's prototype in Fig. 5. Place all components and the PCB in a suitable enclosure and provide terminal strips for the 230V supply input and output separately. The indicator LED and the bypass switch should be positioned on the enclosure cover such that they are visible from the front side. After assembling and test-

ing the timer, the box may be mounted at a suitable place and wired between the supply and the load.

EFY note. The 12V DC relay contacts are rated for maximum current of 6-ampere AC. Therefore, the load (fan/lights) connected to this timer box should be rated lesser than this. **EFY**



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